

Finding the coordinate vector relative to a given basis

Asked 9 years, 10 months ago Modified 5 years, 2 months ago Viewed 230k times



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Given the basis $\beta = \{(1, -1, 3), (-3, 4, 9), (2, -2, 4)\}$ and $x = (8, -9, 6)$, I am to find the corresponding coordinate vector $[x]_\beta$. I claim that the coordinate vectors entries x_1, x_2, x_3 meet the following criterion:



$$x_1(1, -1, 3) + x_2(-3, 4, 9) + x_3(2, -2, 4) = (8, -9, 6)$$



This is equivalent to solving the augmented matrix

$$\begin{bmatrix} 1 & -3 & 2 & 8 \\ -1 & 4 & -2 & -9 \\ 3 & 9 & 4 & 6 \end{bmatrix}$$

which is row equivalent to

$$\begin{bmatrix} 1 & -3 & 2 & 8 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

which gives $x_3 = 0$, $x_2 = -1$ and $x_1 = 3x_2 + 8 = 5$, thus the coordinate vector $(5, -1, 0)$

There is an error here, seeing as the text claims a different answer. What is the error? In particular, is it computational or simply an error in my understanding of the question?

Edit: fix typo in equation

[linear-algebra](#)

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edited Dec 10, 2014 at 21:53



Community Bot
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asked Mar 3, 2014 at 12:43



Andrew Thompson
4,411 6 41 70



in what book you found that exercise? Thanks. – InfZero May 8, 2016 at 19:43

1 Answer

Sorted by: Highest score (default)



Since

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$$5 \cdot \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} + (-1) \cdot \begin{pmatrix} -3 \\ 4 \\ 9 \end{pmatrix} = \begin{pmatrix} 5 \\ -5 \\ 15 \end{pmatrix} + \begin{pmatrix} 3 \\ -4 \\ -9 \end{pmatrix} = \begin{pmatrix} 8 \\ -9 \\ 6 \end{pmatrix},$$



your solution is correct.



Maybe you copied the wrong numbers from your textbook?

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answered Mar 3, 2014 at 12:47



Christoph

24.8k 32 66

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- 6 ▲  Embarrassing, but yes, I did. As a more experienced user: should I delete this question or should I let it stay as a "finding coordinate vectors done right" type of thing? – [Andrew Thompson](#) Mar 3, 2014 at 12:49
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- 7 ▲  It's a perfectly valid question and, most of all, shows your own effort! – [Christoph](#) Mar 3, 2014 at 12:54
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- 4 ▲  @AndrewThompson Thanks for keeping this up :) It was actually helpful to me when learning about coordinate vectors with respect to bases - especially because you didn't make any errors! – [Burt](#) Aug 23, 2020 at 19:42
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